

An exact theory of superconductivity in a Fermi-arc system

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(Dated: December 8, 2025)

$$\hat{H} = \hat{H}_0 + \hat{H}_p, \quad (1)$$

$$\hat{H}_0 = \sum_{\mathbf{k} \in \text{HBZ}} \sum_{\sigma} \hat{H}_{\mathbf{k}\sigma}, \quad \hat{H}_{\mathbf{k}\sigma} = \xi_{\mathbf{k}} \hat{n}_{\mathbf{k}\sigma} + \xi_{\mathbf{k}+\mathbf{Q}} \hat{n}_{\mathbf{k}+\mathbf{Q},\bar{\sigma}} + U \hat{n}_{\mathbf{k}\sigma} \hat{n}_{\mathbf{k}+\mathbf{Q},\bar{\sigma}}, \quad \mathbf{Q} = (\pi, \pi), \quad (2)$$

$$\hat{H}_p = \frac{J}{N} \hat{\Delta}^\dagger \hat{\Delta}, \quad \hat{\Delta} = \sum_{\mathbf{k}} \gamma_{\mathbf{k}} \hat{b}_{\mathbf{k}}, \quad \hat{b}_{\mathbf{k}} = \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow}, \quad \gamma_{\mathbf{k}} = \cos k_x - \cos k_y, \quad (3)$$

$$\chi(\tau) = \frac{\text{Tr}[e^{-\beta \hat{H}} \mathcal{T} \hat{\Delta}_H(\tau) \hat{\Delta}^\dagger]}{N \text{Tr}[e^{-\beta \hat{H}}]}, \quad \chi_0(\tau) = \frac{\text{Tr}[e^{-\beta \hat{H}_0} \mathcal{T} \hat{\Delta}(\tau) \hat{\Delta}^\dagger]}{N \text{Tr}[e^{-\beta \hat{H}_0}]}. \quad (4)$$

Time-dependent operators without subscript H is in the interaction picture evolving under $\hat{H}_0$.

$$\frac{\partial \hat{b}_{\mathbf{k}}(\tau)}{\partial \tau} = e^{\hat{H}_0 \tau} [\hat{H}_0, \hat{b}_{\mathbf{k}}] e^{-\hat{H}_0 \tau}, \quad (5)$$

$$= -2[\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}} + U(\hat{n}_{-\mathbf{k}+\mathbf{Q}\uparrow} + \hat{n}_{\mathbf{k}+\mathbf{Q}\downarrow})] \hat{b}_{\mathbf{k}}(\tau), \quad (6)$$

$$\hat{b}_{\mathbf{k}}(\tau) = e^{-2[\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}} + U(\hat{n}_{-\mathbf{k}+\mathbf{Q}\uparrow} + \hat{n}_{\mathbf{k}+\mathbf{Q}\downarrow})\tau]} \hat{b}_{\mathbf{k}}. \quad (7)$$

$$\chi(\tau) = \frac{X}{Z}, \quad X = \frac{1}{N} \text{Tr}[e^{-\beta \hat{H}_0} \mathcal{T} \hat{S}(\beta, 0) \hat{\Delta}(\tau) \hat{\Delta}^\dagger], \quad Z = \text{Tr}[e^{-\beta \hat{H}_0} \hat{S}(\beta, 0)], \quad \hat{S}(\beta, 0) = \sum_{m=0}^{\infty} \frac{1}{m!} \mathcal{T} \left[\int_0^\beta d\tau' \hat{H}_p(\tau') \right]^m \quad (8)$$

$$X = \sum_{m=0}^{\infty} \frac{1}{Nm!} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_m \text{Tr}[e^{-\beta \sum_{\mathbf{k} \in \text{HBZ}} \sum_{\sigma} \hat{H}_{\mathbf{k}\sigma}} \mathcal{T} \hat{H}_p(\tau_1) \cdots \hat{H}_p(\tau_m) \hat{\Delta}(\tau) \hat{\Delta}^\dagger] \quad (9)$$

$$= \sum_{m=0}^{\infty} \frac{1}{Nm!} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_m \text{Tr} \left[\prod_{\mathbf{k} \in \text{HBZ}} \prod_{\sigma} e^{-\beta \hat{H}_{\mathbf{k}\sigma}} \mathcal{T} \hat{H}_p(\tau_1) \cdots \hat{H}_p(\tau_m) \hat{\Delta}(\tau) \hat{\Delta}^\dagger \right], \quad (10)$$

$$= \sum_{m=0}^{\infty} \frac{J^m}{N^{m+1} m!} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_m \sum_{\mathbf{k}_0 \cdots \mathbf{k}_m} \sum_{\mathbf{q}_0 \cdots \mathbf{q}_m} \gamma_{\mathbf{k}_0} \gamma_{\mathbf{q}_0} \cdots \gamma_{\mathbf{k}_m} \gamma_{\mathbf{q}_m} \\ \times \text{Tr} \left[\prod_{\mathbf{k} \in \text{HBZ}} \prod_{\sigma} e^{-\beta \hat{H}_{\mathbf{k}\sigma}} \mathcal{T} \hat{b}_{\mathbf{k}_1}^\dagger(\tau_1) \hat{b}_{\mathbf{q}_1}(\tau_1) \cdots \hat{b}_{\mathbf{k}_m}^\dagger(\tau_m) \hat{b}_{\mathbf{q}_m}(\tau_m) \hat{b}_{\mathbf{q}_0}(\tau) \hat{b}_{\mathbf{k}_0}^\dagger \right]. \quad (11)$$

$\hat{H}_{\mathbf{k}\sigma}$ commutes with $\hat{b}_{\mathbf{k}'}(\tau)$ except for the coincident cases $\mathbf{k}' = \mathbf{k}, \mathbf{k} + \mathbf{Q}$, which have a vanishing contribution for Eq. (11) as $N \rightarrow \infty$. The trace in Eq. (11) does not vanish only if $\mathbf{k}_0 \cdots \mathbf{k}_m$ and $\mathbf{q}_0 \cdots \mathbf{q}_m$ are the same disregarding their order. There are $m!$ ways to pair $\mathbf{k}_0 \cdots \mathbf{k}_m$ and $\mathbf{q}_0 \cdots \mathbf{q}_m$ to form a connected diagram (consider the sequence $\hat{b}_{\mathbf{q}_0}, \hat{b}_{\mathbf{k}_1}^\dagger, \cdots, \hat{b}_{\mathbf{k}_m}^\dagger, \hat{b}_{\mathbf{k}_0}^\dagger$ and then arranging $\hat{b}_{\mathbf{q}_1}, \cdots, \hat{b}_{\mathbf{q}_m}$ to pair with each of $\hat{b}_{\mathbf{k}_1}^\dagger, \cdots, \hat{b}_{\mathbf{k}_m}^\dagger$ in the sequence), and all these diagrams are the same. Denote $\hat{H}_{\mathbf{k}} = \hat{H}_{\mathbf{k}\uparrow} + \hat{H}_{-\mathbf{k}\downarrow}$. With these, the connected part of Eq. (11) is

$$X_{\text{conn}} = \sum_{m=0}^{\infty} \sum_{\mathbf{k}_0 \cdots \mathbf{k}_m} X_{\text{conn}}(\mathbf{k}_0, \cdots, \mathbf{k}_m), \quad (12)$$

$$X_{\text{conn}}(\mathbf{k}_0, \cdots, \mathbf{k}_m) = \frac{J^m}{N^{m+1}} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_m \gamma_{\mathbf{k}_1}^2 \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}_1}} \mathcal{T} \hat{b}_{\mathbf{k}_1}(\tau) \hat{b}_{\mathbf{k}_1}^\dagger(\tau_1)] \\ \times \gamma_{\mathbf{k}_2}^2 \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}_2}} \mathcal{T} \hat{b}_{\mathbf{k}_2}(\tau_1) \hat{b}_{\mathbf{k}_2}^\dagger(\tau_2)] \cdots \gamma_{\mathbf{k}_0}^2 \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}_0}} \mathcal{T} \hat{b}_{\mathbf{k}_0}(\tau_m) \hat{b}_{\mathbf{k}_0}^\dagger]. \quad (13)$$

$$X = \sum_{m=0}^{\infty} \sum_{\mathbf{k}_0 \cdots \mathbf{k}_m} X_{\text{conn}}(\mathbf{k}_0, \cdots, \mathbf{k}_m) \left\{ \sum_{n=0}^{\infty} \sum_{\mathbf{k}'_1 \cdots \mathbf{k}'_n} X_{\text{disc}}(\mathbf{k}'_1, \cdots, \mathbf{k}'_n) \prod_{\mathbf{k} \notin \{\mathbf{k}_0, \cdots, \mathbf{k}_m, \mathbf{k}'_1 \cdots \mathbf{k}'_n\}} \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}}}] \right\}, \quad (14)$$

$$Z = \sum_{n=0}^{\infty} \sum_{\mathbf{k}'_1 \cdots \mathbf{k}'_n} X_{\text{disc}}(\mathbf{k}'_1, \cdots, \mathbf{k}'_n) \prod_{\mathbf{k} \notin \{\mathbf{k}'_1 \cdots \mathbf{k}'_n\}} \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}}}], \quad (15)$$

$$\chi(\tau) = \frac{X}{Z} = \sum_{m=0}^{\infty} \sum_{\mathbf{k}_0 \cdots \mathbf{k}_m} \frac{X_{\text{conn}}(\mathbf{k}_0, \cdots, \mathbf{k}_m)}{\text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}_0}] \cdots \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}_m}]}]. \quad (16)$$

Similarly and more straightforwardly

$$\chi_0(\tau) = \sum_{\mathbf{k}} \frac{\gamma_{\mathbf{k}}^2 \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}}} \mathcal{T} \hat{b}_{\mathbf{k}}(\tau) \hat{b}_{\mathbf{k}}^{\dagger}]}{N \text{Tr}[e^{-\beta \hat{H}_{\mathbf{k}}}]}, \quad (17)$$

hence

$$\chi(\tau) = \sum_{m=0}^{\infty} J^m \int_0^{\beta} d\tau_1 \cdots \int_0^{\beta} d\tau_m \chi_0(\tau - \tau_1) \chi_0(\tau_1 - \tau_2) \cdots \chi_0(\tau_{m-1} - \tau_m) \chi_0(\tau_m), \quad (18)$$

$$\chi(i\nu_n) = \int_0^{\beta} d\tau e^{i\nu_n \tau} \chi(\tau) = \sum_{m=0}^{\infty} J^m [\chi_0(i\nu_n)]^{m+1} = \frac{\chi_0(i\nu_n)}{1 - J\chi_0(i\nu_n)}. \quad (19)$$

The divergence of $\chi(i\nu_n = 0)$, i.e., $\chi_0(i\nu_n = 0) = 1/J$, marks the superconducting transition point.

From Eq. (17), up to a vanishing contribution from coincident terms with $-\mathbf{k} = \mathbf{k}, \mathbf{k} + \mathbf{Q}$ as $N \rightarrow \infty$,

$$\chi_0(\tau) = \frac{1}{N} \sum_{\mathbf{k}} \gamma_{\mathbf{k}}^2 G_{0\downarrow}(-\mathbf{k}, \tau) G_{0\uparrow}(\mathbf{k}, \tau), \quad (20)$$

where $G_{0\sigma}(\mathbf{k}, \tau) = G_0(\mathbf{k}, \tau)$ is the Green's function of $\hat{H}_0$ and its Fourier component is exactly

$$G_0(\mathbf{k}, i\omega_n) = \frac{1 - n_{\mathbf{k}+\mathbf{Q}}}{i\omega_n - \xi_{\mathbf{k}}} + \frac{n_{\mathbf{k}+\mathbf{Q}}}{i\omega_n - \xi_{\mathbf{k}} - U}, \quad n_{\mathbf{k}+\mathbf{Q}} = \langle \hat{n}_{\mathbf{k}+\mathbf{Q}, \bar{\sigma}} \rangle_0 = \frac{e^{-\beta \xi_{\mathbf{k}+\mathbf{Q}}} + e^{-\beta(\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}} + U)}}{1 + e^{-\beta \xi_{\mathbf{k}}} + e^{-\beta \xi_{\mathbf{k}+\mathbf{Q}}} + e^{-\beta(\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}} + U)}}. \quad (21)$$

$$\chi_0(i\nu_n) = \frac{1}{\beta^2 N} \sum_{\mathbf{k}} \gamma_{\mathbf{k}}^2 \int_0^{\beta} d\tau e^{i\nu_n \tau} \sum_{mm'} e^{-i(\omega_m + \omega_{m'})\tau} G_{0\uparrow}(\mathbf{k}, i\omega_m) G_{0\downarrow}(-\mathbf{k}, i\omega_{m'}), \quad (22)$$

$$= \frac{1}{\beta N} \sum_{\mathbf{k}} \gamma_{\mathbf{k}}^2 \sum_m G_{0\uparrow}(\mathbf{k}, i\omega_m) G_{0\downarrow}(-\mathbf{k}, -i\omega_{m-n}), \quad (23)$$

$$= \frac{1}{N} \sum_{\mathbf{k}} \gamma_{\mathbf{k}}^2 \left\{ \frac{(1 - n_{\mathbf{k}+\mathbf{Q}})^2}{2\xi_{\mathbf{k}} - i\nu_n} \tanh\left(\frac{\beta \xi_{\mathbf{k}}}{2}\right) + \frac{n_{\mathbf{k}+\mathbf{Q}}^2}{2(\xi_{\mathbf{k}} + U) - i\nu_n} \tanh\left(\beta \frac{\xi_{\mathbf{k}} + U}{2}\right) \right. \\ \left. + \frac{n_{\mathbf{k}+\mathbf{Q}}(1 - n_{\mathbf{k}+\mathbf{Q}})}{2\xi_{\mathbf{k}} + U - i\nu_n} \left[\tanh\left(\frac{\beta \xi_{\mathbf{k}}}{2}\right) + \tanh\left(\beta \frac{\xi_{\mathbf{k}} + U}{2}\right) \right] \right\}, \quad (24)$$

$$\chi_0(i\nu_n = 0) = \frac{1}{N} \sum_{\mathbf{k}} \gamma_{\mathbf{k}}^2 \left\{ \frac{1 - n_{\mathbf{k}+\mathbf{Q}}}{2\xi_{\mathbf{k}}} \tanh\left(\frac{\beta \xi_{\mathbf{k}}}{2}\right) + \frac{n_{\mathbf{k}+\mathbf{Q}}}{2(\xi_{\mathbf{k}} + U)} \tanh\left(\beta \frac{\xi_{\mathbf{k}} + U}{2}\right) \right. \\ \left. + \frac{n_{\mathbf{k}+\mathbf{Q}}(1 - n_{\mathbf{k}+\mathbf{Q}})U}{2\xi_{\mathbf{k}} + U} \left[\frac{\tanh(\beta(\xi_{\mathbf{k}} + U)/2)}{2(\xi_{\mathbf{k}} + U)} - \frac{\tanh(\beta \xi_{\mathbf{k}}/2)}{2\xi_{\mathbf{k}}} \right] \right\}. \quad (25)$$

$$= \frac{1}{N} \sum_{\mathbf{k}} \gamma_{\mathbf{k}}^2 \int_{-\infty}^{\infty} d\omega \tilde{A}(\mathbf{k}, \omega) \frac{\tanh(\beta\omega/2)}{2\omega}, \quad (26)$$

with

$$\tilde{A}(\mathbf{k}, \omega) = A(\mathbf{k}, \omega) + A'(\mathbf{k}, \omega), \quad (27)$$

$$A(\mathbf{k}, \omega) = -\Im G_0(\mathbf{k}, \omega + i0^+)/\pi = \sum_{\alpha=l,u} Z_{\mathbf{k}}^{(\alpha)} \delta(\omega - \xi_{\mathbf{k}}^{(\alpha)}), \quad (28)$$

$$= (1 - n_{\mathbf{k}+\mathbf{Q}}) \delta(\omega - \xi_{\mathbf{k}}) + n_{\mathbf{k}+\mathbf{Q}} \delta(\omega - \xi_{\mathbf{k}} - U), \quad (29)$$

$$A'(\mathbf{k}, \omega) = \frac{Z_{\mathbf{k}}^{(l)} Z_{\mathbf{k}}^{(u)} (\xi_{\mathbf{k}}^{(u)} - \xi_{\mathbf{k}}^{(l)})}{\xi_{\mathbf{k}}^{(u)} + \xi_{\mathbf{k}}^{(l)}} [\delta(\omega - \xi_{\mathbf{k}}^{(u)}) - \delta(\omega - \xi_{\mathbf{k}}^{(l)})], \quad (30)$$

$$= \frac{n_{\mathbf{k}+\mathbf{Q}}(1 - n_{\mathbf{k}+\mathbf{Q}})U}{2\xi_{\mathbf{k}} + U} [\delta(\omega - \xi_{\mathbf{k}} - U) - \delta(\omega - \xi_{\mathbf{k}})], \quad (31)$$

where $\xi_{\mathbf{k}}^{(\alpha)}$ and $Z_{\mathbf{k}}^{(\alpha)}$ are the lower ($\alpha = l$) or upper ($\alpha = u$) Hubbard band and its spectral weight, respectively.

$A(\mathbf{k}, \omega)$ is the standard single-particle spectral function of $\hat{H}_0$, whereas $A'(\mathbf{k}, \omega)$ is a many-body correction to the spectral function for the susceptibility. $A'(\mathbf{k}, \omega)$ is proportional to the pseudogap amplitude U when U is small compared to the band width. When the temperature is low enough, $n_{\mathbf{k}+\mathbf{Q}}$ is effectively either zero or one, so $A'(\mathbf{k}, \omega)$ can be neglected and Eq. (26) has the same form as that of a noninteracting electron system. But when the temperature is higher, $0 < n_{\mathbf{k}+\mathbf{Q}} < 1$ for $\mathbf{k}$ near the antinodes (where the pseudogap resides). $\xi_{\mathbf{k}}^{(l)} + \xi_{\mathbf{k}}^{(u)}$ is negative (positive) when μ is closer to the upper (lower) Hubbard band than to the lower (upper) one, so $A'(\mathbf{k}, \omega)$ overall reduces the dominant spectral weight and hence suppresses the superconducting transition temperature.

The summand in the curly brackets of Eq. (25) has no singularity; for $\xi \equiv \xi_{\mathbf{k}} \rightarrow -U/2$ ($n' \equiv n_{\mathbf{k}+\mathbf{Q}}$),

$$\frac{1}{2\xi + U} \left[\frac{\tanh(\beta(\xi + U)/2)}{2(\xi + U)} - \frac{\tanh(\beta\xi/2)}{2\xi} \right] \rightarrow \frac{\beta \operatorname{sech}^2(\beta U/4)}{2U} - \frac{2 \tanh(\beta U/4)}{U^2}, \quad (32)$$

$$\text{Summand} \rightarrow \frac{(1 - n') \tanh(\beta U/4)}{U} + \frac{n' \tanh(\beta U/4)}{U} + n'(1 - n') \left[\frac{\beta \operatorname{sech}^2(\beta U/4)}{2} - \frac{2 \tanh(\beta U/4)}{U} \right], \quad (33)$$

$$= \frac{n'(1 - n')\beta}{2} \operatorname{sech}^2\left(\frac{\beta U}{4}\right) + \frac{n'^2 + (1 - n')^2}{U} \tanh\left(\frac{\beta U}{4}\right). \quad (34)$$

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